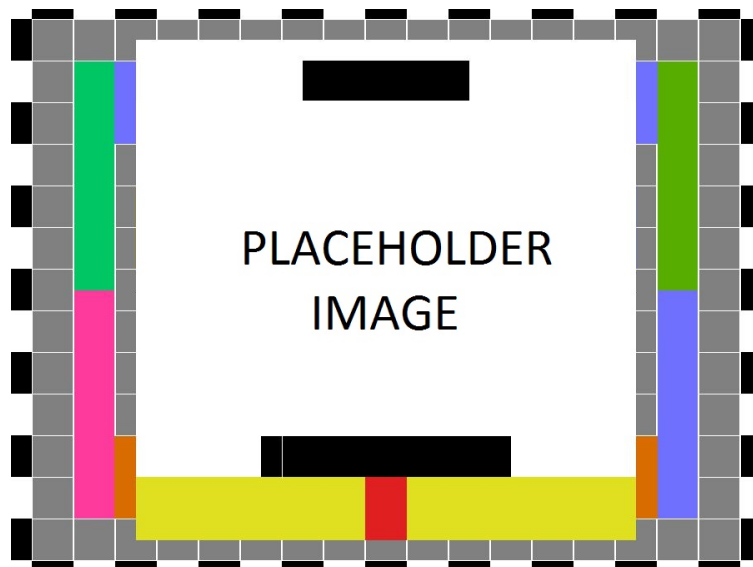


**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
THE UNIVERSITY OF TEXAS AT ARLINGTON**

**DETAILED DESIGN SPECIFICATION
CSE 4317: SENIOR DESIGN II
SPRING 2027**



**TEAM NIMBUS
AQUAGUARD SMART WATER-LEAK MONITOR**

ALAN TURING
GRACE HOPPER
JOHN VON NEUMANN
ADA LOVELACE
CHARLES BABBAGE

REVISION HISTORY

Revision	Date	Author(s)	Description
0.1	01.20.2027	CB	document creation
1.0	02.05.2027	AT, GH, CB	official release

CONTENTS

1	Introduction	4
2	Design Stakeholders & Concerns	4
3	Design Viewpoints	4
4	System Overview	4
5	X Layer Subsystems — Sensor Device	4
5.1	Layer Resources	4
5.2	X.Sensing	4
5.3	X.BLEPeripheral	4
6	Y Layer Subsystems — Mobile Application	4
6.1	Layer Resources	4
6.2	Y.BLECentral	4
6.3	Y.AlertEngine, Y.UI, Y.LocalStore	4
7	Z Layer Subsystems — Cloud Services	5
7.1	Layer Resources	5
7.2	Z.AccountAPI & Z.HistoryStore	5
7.3	Z.PushService	6
8	Requirements Traceability	6
9	Appendix A	6

1 INTRODUCTION

Implementation-level design of AquaGuard Smart Water-Leak Monitor as an IEEE 1016-2009 [1] Software Design Description, building on the SRS and ADS (layers X/Y/Z).

2 DESIGN STAKEHOLDERS & CONCERNS

Firmware/app/cloud implementers, integrators, testers, support; concerns of composition, data, interfaces, timing/state, algorithms, and resources.

3 DESIGN VIEWPOINTS

Context, Composition, Information, Interface, Interaction & State, Algorithm, Resource per IEEE 1016; omitted where not applicable.

4 SYSTEM OVERVIEW

Device: nRF52-class MCU with integrated BLE (C/Zephyr). App: Flutter (Dart) for iOS/Android parity. Cloud: a small serverless API + managed DB + APNs/FCM.

5 X LAYER SUBSYSTEMS — SENSOR DEVICE

5.1 LAYER RESOURCES

Hardware: nRF52 MCU + BLE, capacitive moisture pad, flow sensor, coin/AA battery. **OS:** Zephyr RTOS. **Language:** C.

5.2 X.SENSING

Information: Reading{ts, moisture, flow, battery_mV}. **Algorithm:** periodic ADC sample; a debounced threshold with hysteresis raises leak=true (FR-01); duty-cycled to meet PE-01. **State:** IDLE→SAMPLING→LEAK with cooldown.

5.3 X.BLEPERIPHERAL

Interface (#01, GATT): svc AquaGuard: char reading (notify), leak (notify), threshold (read/write), battery (read). **Interaction:** bonded, encrypted connections only (SE-01); advertises when idle. **Rationale:** standard GATT keeps the app contract simple (AD-01).

6 Y LAYER SUBSYSTEMS — MOBILE APPLICATION

6.1 LAYER RESOURCES

Deps: Flutter, a BLE plugin, a push plugin, local DB (SQLite); **Language:** Dart. **Platforms:** iOS + Android (QA-01).

6.2 Y.BLECENTRAL

Interface (#02): scan/connect/bond; subscribe reading/leak; write threshold. **State:** DISCONNECTED→SCANNING with auto-reconnect. **Rationale:** platform BLE-background differences isolated here so the rest of the app is platform-agnostic (QA-01).

6.3 Y.ALERTENGINE, Y.UI, Y.LOCALSTORE

Y.AlertEngine: on leak while connected, fire local notification/haptic within 3 s (FR-03); if disconnected, POST the event to the backend for push (FR-04). **Y.UI:** guided setup wizard, live status, and history list (US-01, FR-05). **Y.LocalStore:** SQLite events{ts, type, value} and config.

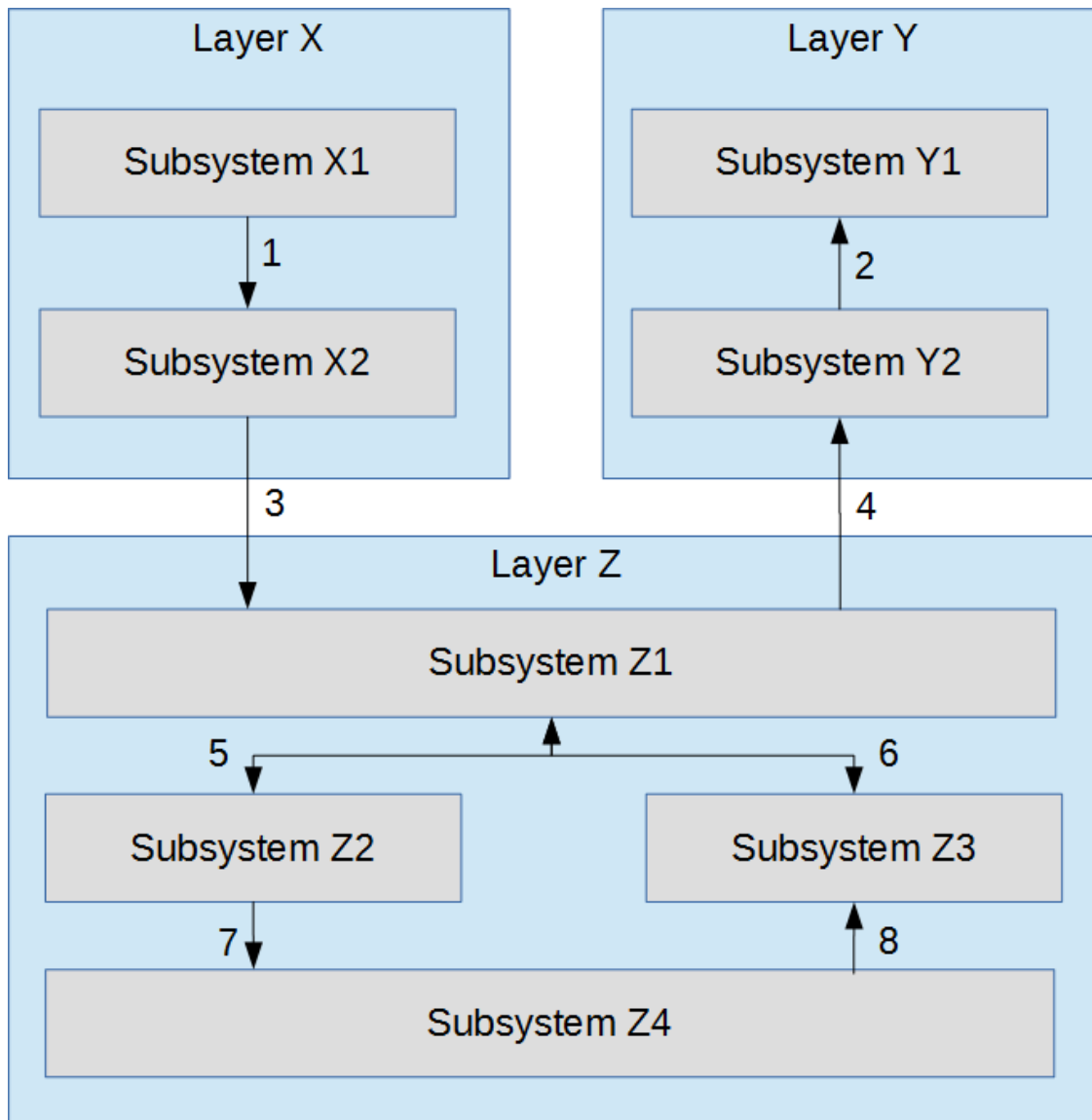


Figure 1: Device/app/cloud architecture carried forward from the ADS

7 Z LAYER SUBSYSTEMS — CLOUD SERVICES

7.1 LAYER RESOURCES

Tech: serverless functions + managed DB; **Language:** TypeScript.

7.2 Z.ACCOUNTAPI & Z.HISTORYSTORE

Interface: POST /events, GET /events, device/user registration over TLS 1.3 (SE-01). **Information:** Event{device_id, ts, type, value} in the managed DB (FR-05).

7.3 Z.PUSHSERVICE

On event ingest, look up the user's device token and send via APNs/FCM (EI-02, FR-04); delivery failures retried and logged (MS-01).

8 REQUIREMENTS TRACEABILITY

Req	Requirement (abbrev.)	Component	Iface
FR-01	Leak detection	X.Sensing	#01
FR-02	Pair/config	Y.BLECentral, X.BLEPeripheral	#02/#01
FR-03	Local alert	Y.AlertEngine	—
FR-04	Remote alert	Z.PushService	—
FR-05	History	Y.LocalStore, Z.HistoryStore	—
SE-01	Secure pair/TLS	X.BLEPeripheral, Z.AccountAPI	#01

9 APPENDIX A

BLE GATT profile table, firmware power budget, enclosure CAD (STEP), and wiring schematic are attached to the closeout package.

REFERENCES

- [1] IEEE, "IEEE Std 1016-2009 — IEEE Standard for Information Technology — Systems Design — Software Design Descriptions," Institute of Electrical and Electronics Engineers, New York, NY, USA, Tech. Rep., 2009.